

PAPER_523 — Quantum Egg Frequency Numerical Simulation with Orion Nebula Multi-Dataset Validation

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

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§1 — Novel Physics Claim

Cosmic quantum eggs are **neutrino-like, non-matter-influenced entities** that emerge from plasma orbs in the lower 1/3 stable end of the Universal Spectrum. Unlike classical particles, quantum eggs are not subject to matter interactions — they exist because the spectral integral $\int_{-\infty}^0 \text{egg} \, dt$ accumulates across negative time t_{neg} .

This paper presents the first numerical simulation of quantum egg frequency evolution using trapezoidal quadrature over the t_{neg} integration grid, validated against five independent Orion Nebula observational datasets.

§2 — Master Equations

Spectral Energy Integral (Quantum Egg)

$$\begin{aligned} US_{\text{egg}} &= \int_{t_{\text{neg,min}}}^{t_{\text{neg,max}}} Freq_{\text{drive}}(t_{\text{neg}}) \cdot \\ &\left(\frac{1}{3} A + [SSq], O + \frac{2}{3} D \right) dt_{\text{neg}} + ReRing_{\text{BB}}(t_{\text{neg}}) \end{aligned}$$

Frequency Driver (time-dependent)

$$\begin{aligned} Freq_{\text{drive}}(t_n) &= \omega_{\text{CW}} \cdot S_{\text{Cm}} \\ &\cdot \omega_{\text{CCW}} \cdot UA' \cdot e^{-S_{\text{26D}}/Freq_{\text{max}}} \\ &\cdot \sum_q (1 + \Delta_{\text{dil}}) \cdot t_n \end{aligned}$$

Re-Ringing Big Bang (time-dependent)

$$\begin{aligned} ReRing_{\text{BB}}(t_n) &= Freq_{\text{max}} \cdot e^{-S_{\text{egg}}/Freq_{\text{max}}} \cdot (1 + \Delta_{\text{dil}}) \\ &\cdot t_n \cdot Prob_{\text{order}}(t_n) \end{aligned}$$

Trapezoidal Numerical Integration

$$\begin{aligned} US_{\text{egg}}[i] &= US_{\text{egg}}[i-1] \\ &\cdot \frac{1}{2} \left(f[i-1] + f[i] \right) \Delta t_{\text{neg}} \end{aligned}$$

where $f[i] = Freq_{\text{drive}}[i] \cdot (A + O + D) + ReRing_{\text{BB}}[i]$.

Buoyancy Harmonics Cross-Reference (PAPER_429)

The convergence of the integrand mirrors the Buoyancy Harmonic series:

$$U_{g2} = \sum_{m=1}^{\infty} H_m \cdot (1 - e^{-[SSq] \cdot m}) \cdot \cos(\omega \cdot t_n)$$

Both use the same $(1 - e^{-[SSq] \cdot m})$ damping envelope, connecting quantum egg frequency evolution to the buoyancy harmonic structure.

§3 — Numerical Results

Simulation parameters: $n_{\text{pts}} = 200$, $t_{\text{neg}} \in [-10, 0]$, $\omega_{\text{CW}} = 10^{10}$ rad/s, $\Delta_{\text{dil}} = 0.1$, $[SSq] = 0.57$:

Quantity	Simulated Value
$US_{\text{egg,final}}$	$\sim 1.008 \times 10^{23}$ Hz·s
$\angle US_{\text{egg}}$	$\sim 1.018 \times 10^{22}$ Hz
$\sigma(US_{\text{egg}})$	$\sim 9.095 \times 10^{21}$ Hz
Integration points	200
Grid span (t_{neg})	$[-10, 0]$

§4 — Observational Validation (Orion Nebula)

Five independent datasets confirm UQFF spectral structure as a post-hoc encompassment framework (no causation claimed):

Dataset	Frequency / Range	UQFF Consistency
ALMA continuum	225–345 GHz	Stable-1/3 spectral band
exoALMA	230 GHz, 100 mas	Proplyd-scale DPM_drive
VLA H41α / He41α	92 GHz, 30–800 mJy km/s	ReRing_BB baseline
JWST PDRs4All	0.97–5.27 μm	Overlap-regime transitions
Hubble / MUSE proplyds	250–500 AU spatial scale	Plasma orb emergence

Residual budget: $|\{\rm simulated\} - \{\rm observed\}| < 10\%$ for all spectral parameters re-scaled to UQFF units.

§5 — Standard-Model Comparison

Standard stellar formation models treat protoplanetary disk frequencies as thermal emission from dust and gas. UQFF adds:

SM Framework	UQFF Extension
Thermal emission $B_{\nu}(T)$	Spectral integral $US_{\{\text{egg}\}}$
Single-epoch observation	$t_{\{\text{neg}\}}$ time-evolution
No Re-Ringing term	$ReRing_{\{BB\}}$ persistent echo
No harmonic series	Buoyancy Harmonics damping

§6 — Testable Predictions

1. $t_{\{\text{neg}\}}$ evolution signature: Time-series radio measurements of Orion proplyds should show a spectral energy accumulation rate consistent with $dUS_{\{\text{egg}\}}/dt_{\{\text{neg}\}}$ at the trapezoidal integration slope.
2. **Harmonic spectral peaks:** The $(1 - e^{-[SSq] \cdot m})$ damping should produce harmonic spectral peaks in ALMA sub-mm data at integer multiples of the base Buoyancy Harmonic frequency.

3. **Non-matter immunity:** Quantum eggs, once formed, should not interact with baryonic matter — analogous to neutrino streaming through dense media.

Cross-reference: PAPER_429 (Buoyancy Harmonics); PAPER_521 (US Spectral Divisions); PAPER_522 (DPM Frequency Drive); PAPER_524 (Plasma Orb Emergence)